

Esercizio 1. Let $A \models \mathcal{T}$ and denote by $pp(A)$ the set of pp formulas with parameters from A . Consider the category $Mod(\mathcal{T})$ (with arrows morphisms between models of $\mathcal{T}$). Then one can construct an ec model of $\mathcal{T}$ as a direct limit starting from A .

Proof. Let $\kappa = |pp(A)|$ and let $e : \kappa \rightarrow \kappa \times \kappa$ be some dove-tail enumeration. We will now construct a direct system $\{A_\alpha : \alpha < \kappa\}$ with direct arrows $f_{i,j}$ for $i \leq j < \kappa$. Let $PP = \{\phi_{\alpha,\beta} : \alpha, \beta < \kappa\}$ where $\phi_{\alpha,\beta}$ is the α -th formula enumerated in $pp(A_\beta)$. Consider the following direct system:

- $A_0 = A$
- $A_{\alpha+1}$ is some model of $\mathcal{T} \cup Diag^+(A_\alpha) \cup \{\phi_{e(\alpha)}\}$ if consistent and A_α otherwise
- A_β is the direct limit of $\{A_\alpha : \alpha < \beta\}$ if β limit

As direct arrows, take the ones given by $Diag^+(A_\bullet)$. So for example: if $i < j$, by construction we have $Diag^+(A_0) \subseteq Diag^+(A_1) \subseteq \dots \subseteq Diag^+(A_i) \subseteq \dots \subseteq Diag^+(A_j)$. Therefore including fresh constants at each steps the direct arrows are determined.

It is clear that this is a direct system, but we need to show that such a direct system exists. In particular we claim that direct limit exist. Given a direct system $(M_i : i < \lambda)$ with direct arrows $(f_{i,j} : i \leq j < \lambda)$ we build the direct system as the following structure: Put $m \equiv n$ if there are $i \leq j < \lambda$ such that $f_{i,j}(m) = n$ (or viceversa) to generate an equivalence relation on $\bigsqcup_{i < \lambda} M_i$.

- $M = \{[m] : m \in M_i \text{ for some } i < \lambda\}$
- $M \models F([m]) = [n]$ iff $\exists i < \lambda \exists m', n' \in M_i \ m' \equiv m \wedge n' \equiv n \wedge M_i \models F(m') = n'$.
- $M \models R([m])$ iff $\exists i < \lambda \exists m' \in M_i \ m \equiv m' \wedge M_i \models R(m')$.

We need to prove:

- M is well defined: by contradiction let's say $\exists i \leq j \exists m_1, n_1 \in M_i \exists m_2, n_2 \in M_j \ m_1 \equiv m_2 \wedge n_1 \equiv n_2 \wedge M_i \models F(m_1) = n_1 \not\iff M_j \models F(m_2) = n_2$. For example we may suppose $i < j$, $M_i \models F(m_1) = n_1$, $M_j \models F(m_2) \neq n_2$. This means $M_j \models F(f_{i,j}(m_1)) \neq f_{i,j}(n_1)$ which is a contradiction since $f_{i,j}$ is a morphism. One treats other cases similarly.
- $M \models \mathcal{T}$: By contradiction, assume $M \models \phi$ where ϕ is a pp formula such that $\neg\phi \in \mathcal{T}$. Then ϕ will be of the form $\exists x_1, \dots, x_n \bigwedge Q(x_1, \dots, x_n)$. Let $[m_1], \dots, [m_n] \in M$ be such that $M \models \bigwedge Q([m_1], \dots, [m_n])$. Let $\alpha < \lambda$ such that $[m_1], \dots, [m_n] \in M_\alpha$. By definition of M we have $M_\alpha \models \bigwedge Q(m_1, \dots, m_n)$. So $M_\alpha \models \phi$. This is a contradiction since by construction $M_\alpha \models \neg\phi$.

- M is the direct limit: Let $E \models \mathcal{T}$, $(g_i : M_i \rightarrow E)_{i < \lambda}$ arrows such that $\forall i \leq j < \lambda$ $g_j \circ f_{i,j} = g_i$. Now consider $h : M \rightarrow E, [m] \mapsto g_i(m)$. This is well defined: say $[m_i] = [m_j]$ with $m_i \in M_i, m_j \in M_j$ and $i < j$ (the case $i = j$ can be treated using the same argument two times taking a third index $l > i = j$). Then $m_j = f_{i,j}(m_i)$ so we need to prove that $g_j(f_{i,j}(m_i)) = g_i(m_i)$, which is true by hypothesis.

Now we only need to prove that A_κ (the direct limit of the whole direct system):

- is a model of $\mathcal{T}$: ok from observations above
- is direct limit: ok by definition (the existence is ok from above)
- is ec: by contradiction, let $f : A_\kappa \rightarrow B$ morphism for some $B \models \mathcal{T}, \phi(f(a))$ with $A_\kappa \not\models \phi(a)$ and ϕ pp formula. Let $\alpha < \kappa$ such that $\phi(a) = \phi_{e(\alpha)}$. Since $A_\alpha \rightarrow A_\kappa \rightarrow B$ we also have $A_\alpha \rightarrow B$, so $B \models \text{Diag}^+(A_\alpha)$. In particular $\mathcal{T} \cup \text{Diag}^+(A_\alpha) \cup \{\phi_{e(\alpha)}\}$ is consistent. Therefore $A_{\alpha+1} \models \phi_{e(\alpha)}$ and consequently $A_\kappa \models \phi_{e(\alpha)}$, contradiction.

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